

# OPERATING SYSTEMS CONCEPT – LAB EXP – 08:

## BANKERS DEADLOCK AVOIDANCE ALGORITHM

```
Banker's Algorithm in C
Apr 20 17:39
snoordeen@localhost:~/os-lab/bankers — vim bankers.c

1 #include <stdio.h>
2 #define MAX 10
3 int main() {
4     int m, n, i, j, k, need;
5     printf("Enter number of processes: ");
6     scanf("%d", &m);
7     printf("Enter number of resources types: ");
8     scanf("%d", &n);
9     int available[n], max[n][n], allocation[n][n], need[n][n];
10    int finish[n] = {0}, safe_seq[MAX], count = 0;
11
12    // Allocation matrix
13    printf("Enter allocation matrix:\n");
14    for (i = 0; i < m; i++) {
15        for (j = 0; j < n; j++) {
16            scanf("%d", &allocation[i][j]);
17        }
18    }
19    // Max matrix
20    printf("Enter max matrix:\n");
21    for (i = 0; i < m; i++) {
22        for (j = 0; j < n; j++) {
23            scanf("%d", &max[i][j]);
24        }
25    }
26    // Calculate need matrix
27    printf("Need matrix:\n");
28    for (i = 0; i < m; i++) {
29        for (j = 0; j < n; j++) {
30            need[i][j] = max[i][j] - allocation[i][j];
31            printf("%d ", need[i][j]);
32        }
33        printf("\n");
34    }
35    // Available resources
36    printf("Enter available resources:\n");
37    for (j = 0; j < n; j++) {
38        scanf("%d", &available[j]);
39    }
40    // Banker's Algorithm
41    printf("\nSafe sequence is:\n");
42    while (count < m) {
43        int found = 0;
44        for (i = 0; i < m; i++) {
45            if (finish[i] == 0) {
46                int j;
47                for (j = 0; j < n; j++) {
48                    if (need[i][j] > available[j])
49                        break;
50                }
51                if (j == n) {
52                    for (k = 0; k < n; k++)
53                        available[k] += allocation[i][k];
54                    safe_seq[count++] = i;
55                    finish[i] = 1;
56                    found = 1;
57                    printf("P%d ", i);
58                }
59            }
60        }
61        if (!found) {
62            printf("\nSystem is in unsafe state! Deadlock may occur.\n");
63            return 0;
64        }
65    }
66    printf("\nSystem is in safe state.\n");
67    return 0;
68 }
```

```
Apr 20 17:39
snoordeen@localhost:~/os-lab/bankers
snoordeen@localhost:~/os-lab/bankers$ gcc bankers.c -o bankers
snoordeen@localhost:~/os-lab/bankers$ ./bankers
Enter number of processes: 3
Enter number of resources types: 3
Enter allocation matrix:
0 1 0
2 0 0
3 0 2
Enter max matrix:
7 5 3
1 2 2
3 0 2
Need matrix:
7 4 3
1 2 2
0 0 0
Enter available resources:
3 3 2
Safe sequence is:
P0 P2
System is in safe state.
snoordeen@localhost:~/os-lab/bankers$
```

```
Activities Terminal Apr 20 17:28
snoordeen@localhost:~/os-lab/bankers

snoordeen@localhost:~/os-lab/bankers$ vi bankers.c
snoordeen@localhost:~/os-lab/bankers$ gcc bankers.c -o bankers
snoordeen@localhost:~/os-lab/bankers$ ./bankers
Enter number of processes: 5
Enter number of resource types: 3
Enter allocation matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Max matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter available resources:
3 3 2
Need matrix:
7 4 3
1 2 2
6 0 0
0 1 1
4 3 1
P1 executed. Available now: 5 3 2
P3 executed. Available now: 7 4 3
P4 executed. Available now: 7 4 5
P0 executed. Available now: 7 5 5
P2 executed. Available now: 10 5 7
Safe sequence is: P1 P3 P4 P0 P2
snoordeen@localhost:~/os-lab/bankers$
```